UMBRA

The autonomous circuit breaker for DeFi

Detect systemic pool stress — and evacuate capital before liquidity evaporates.

COMPOSITE BREAK-RISK INDEX · CBRI V1.0

ETH Global Lisbon 2026 · github.com/nadaamd/umbra

THE PROBLEM

DeFi breaks faster than people react.

When a pool goes into crisis, price falls and depth collapses at once.

By the time you understand what's happening, the liquidity that would have let you exit cheaply is gone — and your sell order eats the wall.

$60B+	**~$0.87**	**billions**
Terra / UST	USDC depeg low	lost — mostly to late reactions

THE IDEA

One score. 0 → 100. One trigger.

CBRI is a continuous break-risk score, updated every 5 minutes.

Cross the proven threshold τ^* → Umbra auto-evacuates the position to a safe stablecoin.

Three orthogonal tells

01

Liquidity-drain velocity

$$v_L = -\frac{1}{L}\frac{\Delta L}{\Delta t}$$

LPs pulling out — and accelerating. Leads by hours.

EARLY WARNING

02

Order-flow imbalance

$$I = \frac{|\sum \text{in} - \sum \text{out}|}{\sum |\text{flow}|}$$

How one-directional the flow is. The rush to exit.

THE RUSH

03

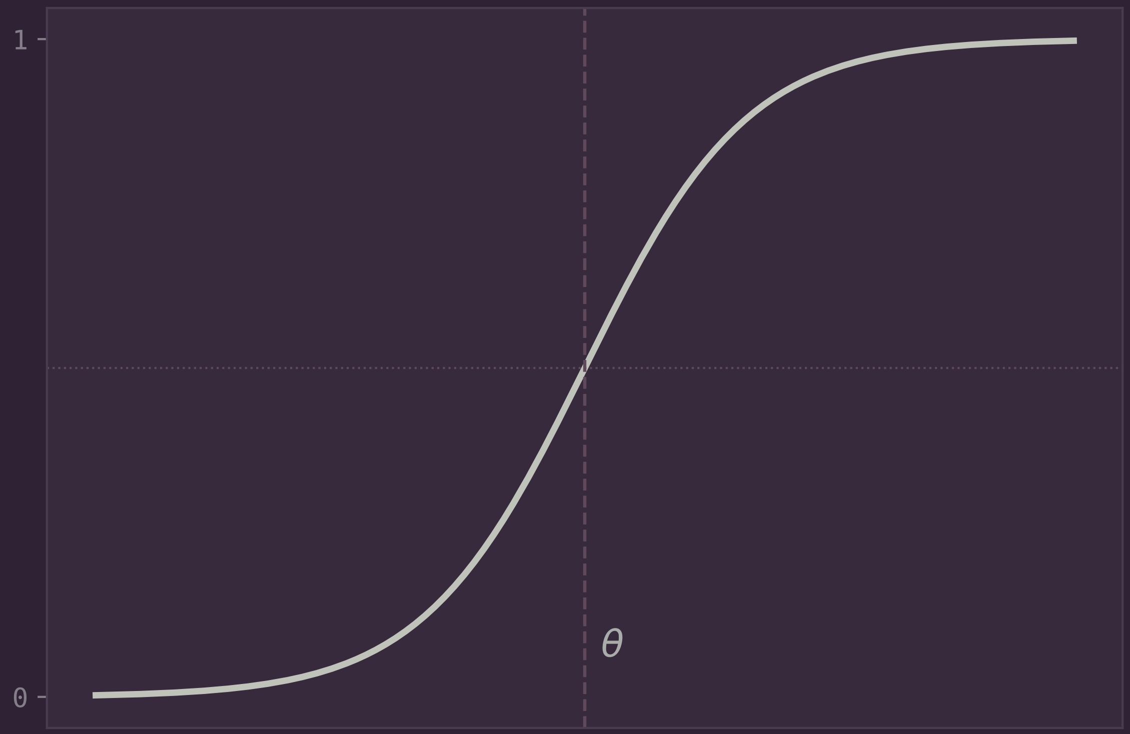
Peg divergence

$$D = |1 - P|$$

The price itself leaving its $1 peg. Unambiguous.

CONFIRMATION

A soft, tunable threshold



$$s_i = \sigma(\alpha_i\,(x_i - \theta_i))$$

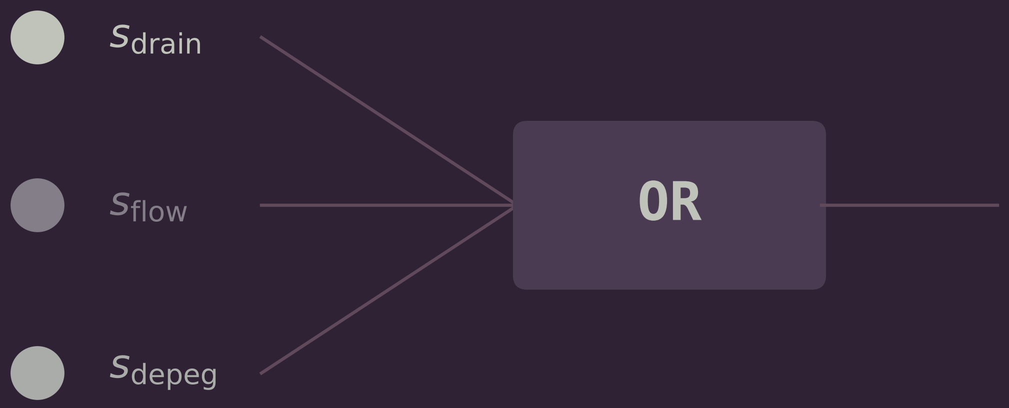
Each raw signal → a common [0,1] axis.

θ = where it starts to look abnormal.

α = how sharply it flips.

Read it as: the probability, through this one lens, that the pool is breaking.

A breaker trips on ANY red



CBRI → 100

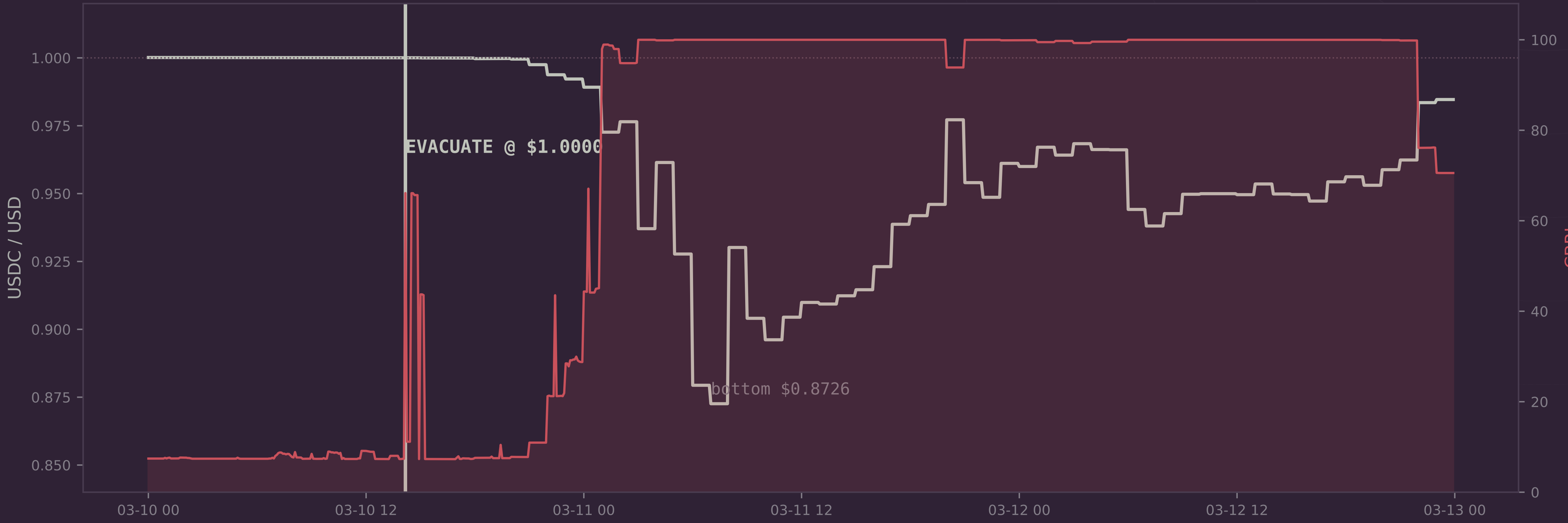
$$\mathrm{CBRI} = 100\left(1 - \prod_i (1 - w_i s_i)\right)$$

Not an average — one extreme signal is enough. Non-informative signals get weight 0 and vanish.

Calm reads ~8. Crisis reads 100.

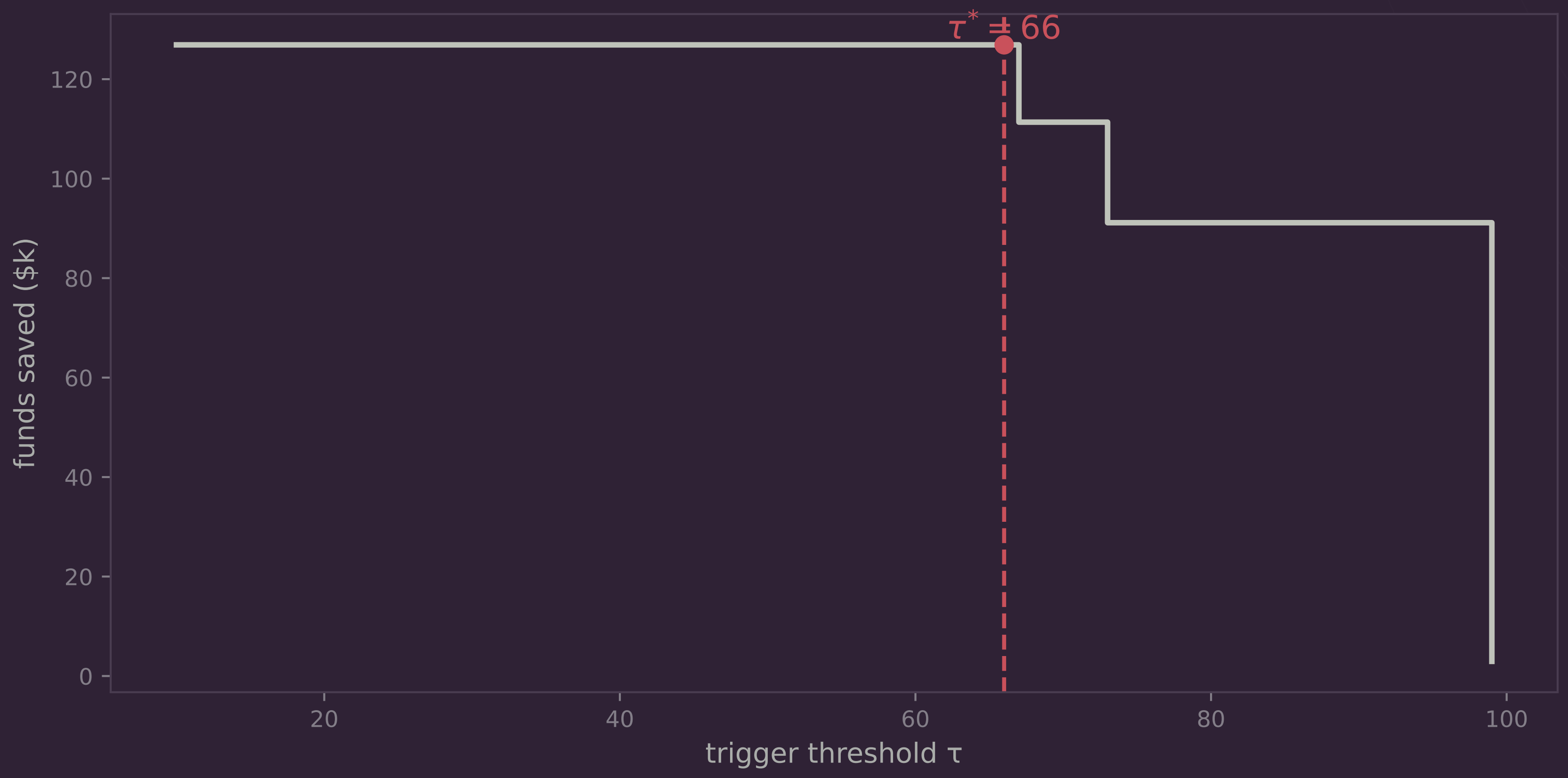
Thresholds are set so each sigmoid rests near zero and saturates only in genuine stress — a clean separation with zero false alarms in the quiet window.

17 hours of early warning



$126,900 saved

on a $1M position — 12.7%



τ* = 66

highest safe threshold

+$126.9k

vs suffering the bottom

5 bps

exit slippage at τ*

−$124k

cost of waiting to τ=99

Liquidity vanishes ×23,000,000

As USDC leaves the peg, the active depth of the USDC/USDT pool collapses by seven orders of magnitude — so a fixed $1M exit hits a wall.

5 bps

exit at τ^* — near free

→

657 bps

exit at $\tau=99$ — the wall

The stack

The Graph

DATA BACKBONE · TARGET TRACK

Tick-level swaps, mints & burns across pools — 48,066 real events in seconds.

0G

VERIFIABLE ON-CHAIN AI · TARGET TRACK

The risk model + every score anchored on decentralized storage. Auditable, tamper-evident.

Uniswap v3

EXECUTION · INFRA

Live on-chain quote (QuoterV2) + swap (SwapRouter02) for the evacuation. Real depth, no API.

B U S I N E S S   M O D E L

We earn only on the loss we avoid.

A success fee charged only on funds saved.

On this event: 10% = 12, 690—*earnedonlybecausetheuserkept*126,900.

W E   W I N   I F   A N D   O N L Y   I F   T H E   U S E R   W I N S .

The Graph · 0G · Uniswap v3   A working, tested, on-chain circuit breaker for DeFi.